

WALTER STEVE POLLARD JR.

SENIOR SOFTWARE ENGINEER · VIDEO · CHROMECAST · HLS/DASH · TYPESCRIPT · JAVASCRIPT · PHILADELPHIA, PA (OPEN TO REMOTE)

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SUMMARY

I'm a Senior Software Engineer with 25 years specializing in web video player development. I've been involved with video on the web since the early Flash player days, Macromedia RTMP/E, the start of FLV video, and have always stayed in the video space. Over the years, I've built all kinds of video experiences, with debugging tooling for support and internal analytics, streaming performance optimization, and media delivery at scale.

I'm a 13-year Subject Matter Expert for Xfinity Stream's HD video player at Comcast, the desktop and Chromecast video player serving millions to watch TV, sports, and on-demand content. Deep expertise in HLS, MPEG-DASH, Google Cast Application Framework (CAF), dynamic ad insertion (DAI/SSAI/CSAI via FreeWheel, Google IMA SDK), DRM-protected playback (Widevine, PlayReady), CDN delivery (Akamai), and custom debugging tooling. On every team I've joined, I became the engineer others came to: mentoring developers, leading code reviews, and building the shared tooling and APIs the organization relied on. I'm a C.O.D.A. (Child of Deaf Adults) and my first language is ASL, so web accessibility is personal for me. Currently building AI-augmented media experiences with Next.js 15, TypeScript, and Claude Code.

SKILLS

Languages TypeScript · JavaScript · HTML5 · CSS3

Frameworks & Libraries React · Next.js · Web Components · Lit · Google Polymer · Tailwind CSS · Video.js · Dash.js

Video & Streaming HLS · MPEG-DASH · MSE · EME · CMAF · fMP4 · Google Cast CAF · multi-bitrate ABR · Widevine · PlayReady · DRM-protected playback

Ad Tech DAI · SSAI · CSAI · VAST 3/4 · VMAP · FreeWheel · Google IMA SDK

Engineering Practices REST API Design · Code Review · Mentoring & Team Enablement · System Design · Performance Optimization · Cross-functional Collaboration · QA Collaboration · Unit & Integration Testing (Jest) · CI/CD

Tooling & Infra Git · GitHub Actions · CI/CD pipelines · Webpack · ESLint · Docker · Akamai CDN

AI Tools Claude Code · GitHub Copilot · Google Gemini · AI-assisted engineering workflows

EXPERIENCE

Senior Software Engineer 4

Comcast Technology Solutions, Philadelphia, PA *June 2012 – June 2025 (13 years)*

Subject Matter Expert for Xfinity Stream's HD video player on desktop and Chromecast, serving at national cable scale with millions of concurrent viewers.

- Built and maintained HD video player (HLS/DASH/MSE/EME) for Xfinity Stream desktop and Chromecast
 - Led Chromecast product from MVP to production; worked directly with Google's Cast Application Framework team
 - Designed and built the Google Cast CAF sender/receiver API powering Chromecast playback for both desktop and mobile cast clients
 - Implemented DAI/SSAI/CSAI dynamic ad insertion for VOD, SVOD, and live streams using FreeWheel and Google IMA SDK
 - Integrated Adobe Primetime PSDK for DRM-protected content delivery (Widevine, PlayReady)
 - Built custom debugging tooling and internal REST APIs used across the engineering organization for HD video playback diagnostics
 - Worked daily with QA, folding their user-testing feedback into the player and building the diagnostics they tested with; treated unit testing as a baseline for anything shipping at scale
 - Served as the org-wide SME and reviewer for HD video playback, mentoring engineers on every team I joined and shaping shared front-end and debugging standards
 - Wrote Chromecast device detection code enabling generational device logging across Comcast subscriber base
 - Enabled multipartner support across 5 Comcast cable partners
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Sr. Front End Developer (Consulting → Full-Time)

FIS / Sungard consulting at **Comcast Interactive Media**, Philadelphia, PA *February 2012 – 2017*

Consulting engagement that led to full-time SSE4 hire. Built Google Polymer / Web Components front-end for one of the highest-traffic video platforms in the country with 5M+ monthly views.

- Google Polymer framework adoption and Web Components architecture
 - Adobe Primetime PSDK video player integration
 - JavaScript front-end architecture at scale
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Freelance Sr. Flash Developer

Constellation.tv, New York City *August 2011 – December 2011*

Built a live HD group movie-watching experience with webcam support, people watching films together in real time. OSMF player on Akamai's full CDN stack.

- OSMF player on Akamai RTMP/HDN1/HDN2/SecureHD
 - Live HD video + web camera capture and broadcast (Flash NetStream)
 - Adobe Flash Media Server 4.x live stream ingest and relay
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Sr. Flash Developer

Empathy Lab, Conshohocken, PA *September 2010 – November 2011*

AS3/Flex development for major media clients. Built UltraViolet DRM player and Rogers OnDemand adaptive streaming player.

- Rogers OnDemand player with Akamai HDCore adaptive CDN
 - UltraViolet (UVU) DRM-protected cloud playback
 - Android 2.2+ native video player
 - Sprout Online children's platform
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Sr. Flash Developer

Razorfish Health, Philadelphia, PA *December 2009 – March 2011*

AS3 front-end development with server-side data feed integrations for healthcare clients.

Freelance Full-Stack Developer

Sierra Star Studio, Independent *July 2009 – June 2012*

Ran freelance practice alongside contracted roles. Full-stack web development for various clients.

Flash Developer, E! Online / Comcast Entertainment Group

Los Angeles, CA · *December 2008 – August 2009* · Flash video and ad development for E! Online platform.

Sr. Flash Developer & Technology Leader, FEARnet / Comcast Entertainment Group

Los Angeles, CA · *September 2007 – January 2009* · Led front-end development for FEARnet.com. Brightcove video integration, DoubleClick DART ad serving, Omniture analytics. Orchestrated site migration to Comcast Entertainment Group infrastructure.

Flash Developer, Vantage Inc

New Hope, PA · *May 2004 – January 2006* · Flash/ActionScript development, interactive experiences and assessment applications.

Web Developer / Systems Test Engineer, Unisys

Malvern, PA · *September 1999 – December 2002* · HTML, JavaScript, web development and systems testing while completing EECS coursework at Drexel University.

EDUCATION

B.S. Electrical Engineering / Computer Science (coursework) Drexel University, Philadelphia, PA | 1997 – 2001

Network Systems Professional Certification Cittone Institute | 2001 – 2002

CURRENTLY BUILDING

wsp-cast-sender / wsp-cast-receiver Personal project: Next.js 15 Chromecast Sender + Receiver portfolio application

- Next.js 15 · TypeScript · React 18 · Google Cast SDK / CAF
- Video.js · Dash.js · ReactPlayer (HLS / MPEG-DASH / Twitch)
- GitHub Actions CI/CD · Static export · SSH/SCP production deploy
- Live at walter.pollardjr.com | github.com/NooRotic/wsp-cast-sender

RipTheLeetBot AI-powered interview prep trainer, Electron desktop app

- Electron 33 · React 18 · TypeScript · xterm.js · Google Gemini
- github.com/NooRotic/RipTheLeetBot

JobTrackr Developer-first job search dashboard

- SvelteKit 5 · Svelte 5 · Tailwind CSS v4 · Bun · TypeScript
- github.com/NooRotic/JobTrackr